

Fake News Detection System

A full-stack fake news detection platform that analyzes text, URLs, and images using a multi-layer pipeline:

- ML layer (DistilBERT multilingual model)
- LLM verification layer (Gemini/OpenAI/Ollama fallback)
- Fusion layer (final confidence + risk decision)
- OCR for image-based news analysis
- MongoDB persistence for user profile, history, and analytics

This repository contains both backend (Flask/Python) and frontend (React/Vite/TypeScript).

1. What This Project Does

The system supports:

- Text verification
- URL/article verification (with scraping)
- Image verification (OCR + analysis)
- Claim verification
- Per-user profile/history/analytics data isolation
- Account operations (profile save, password change, delete account)

2. High-Level Architecture

Backend

- Framework: Flask
- Main entry: `backend/app.py`
- Core services: `backend/services/`
 - `fake_news_service.py` (ML model inference)
 - `openai_verification_service.py` (LLM verification)
 - `fusion_service.py` (evidence fusion)
 - `ocr_service.py` (Tesseract OCR)
 - `mongodb_service.py` (MongoDB operations)
 - `cloudinary_service.py` (optional image upload)
- Utilities: `backend/utils/`

Frontend

- Framework: React + Vite + TypeScript
- Main app: `frontend/src/`
- API client: `frontend/src/services/api.ts`
- Auth/session helper: `frontend/src/services/auth.ts`
- Pages: dashboard, verify-text, verify-url, verify-image, history, analytics, profile, login, signup

Data Stores

- MongoDB Atlas (or compatible MongoDB)
 - Profiles ([user_profiles](#))
 - Predictions ([predictions](#))
 - Analytics aggregation

3. Project Structure

```
fake new/  
  backend/  
    app.py  
    config.py  
    requirements.txt  
    requirements-training.txt  
    services/  
    utils/  
    model/  
    data/  
    training/  
    .env.example  
  frontend/  
    package.json  
    src/  
    .env.example  
  README.md  
  QUICK_START.md  
  start-dev.ps1
```

4. Prerequisites

- Python 3.10+ (3.10.x recommended in this workspace)
- Node.js 18+
- npm 9+
- Tesseract OCR installed and available in PATH
- MongoDB Atlas connection string (recommended for full features)

5. Backend Setup

From repository root:

```
cd backend
```

Option A: Use existing local environment ([venv310](#))

```
.\venv310\Scripts\python.exe -m pip install -r requirements.txt
```

Option B: Create a new virtual environment

```
python -m venv .venv
.\.venv\Scripts\activate
pip install -r requirements.txt
```

Configure environment

Create and edit `.env`:

```
copy .env.example .env
```

Important variables (example):

```
MONGODB_URI=your_mongodb_connection_string
GEMINI_API_KEY=your_key_optional
OPENAI_API_KEY=your_key_optional
NEWSAPI_KEY=your_key_optional
COINGECKO_ENABLED=true
TESSERACT_CMD=C:\Program Files\Tesseract-OCR\tesseract.exe
```

Run backend server

If using `venv310`:

```
.\venv310\Scripts\python.exe app.py
```

Backend default URL:

- `http://localhost:5000`

6. Frontend Setup

From repository root:

```
cd frontend
npm install
```

Create frontend `.env`:

```
VITE_API_URL=http://localhost:5000/api
```

Run dev server:

```
npm run dev
```

Frontend default URL:

- <http://localhost:3000> (or Vite-provided port)

7. API Endpoints

System

- `GET /api/health`
- `GET /api/model-metrics`

Prediction

- `POST /api/predict-text`
- `POST /api/predict-url`
- `POST /api/predict-image`
- `POST /api/verify-claim`
- `POST /api/source-credibility`

User/Account

- `GET /api/profile`
- `POST /api/profile`
- `POST /api/change-password`
- `POST /api/delete-account`

History/Analytics

- `GET /api/history`
- `GET /api/analytics`
- `GET /api/stats`

Note: user-scoped endpoints expect `user_email` context from frontend.

8. Prediction Flow (Simplified)

Text flow

1. Validate input
2. Detect language
3. ML prediction
4. LLM verification
5. Fusion scoring

6. Explanation generation
7. Save prediction to MongoDB
8. Return final result

URL flow

1. Scrape URL content
2. Detect language
3. ML prediction on extracted article text
4. LLM URL verification
5. Fusion + explanation
6. Save to MongoDB

Image flow

1. Upload image
2. OCR text extraction
3. Language detection
4. ML + LLM image/text verification
5. Fusion + explanation
6. Optional Cloudinary upload
7. Save to MongoDB

9. Account and Data Behavior

- Each account is identified by user email.
- Profile, history, analytics, and stats are user-scoped.
- New account starts with empty history/stats until that user makes predictions.
- Password changes are persisted in MongoDB with bcrypt hashing.
- Delete account removes user profile and user-scoped predictions.

10. Common Troubleshooting

Backend fails with `ModuleNotFoundError`

Install dependencies in the same interpreter used to run app:

```
cd backend
.\venv310\Scripts\python.exe -m pip install -r requirements.txt
```

Error: `Cannot find path ...\backend\backend`

You are already inside `backend`. Use:

```
.\venv310\Scripts\python.exe app.py
```

MongoDB DNS errors (`getaddrinfo failed`)

- Check internet and DNS stability
- Ensure MongoDB Atlas network access allows your IP
- Try DNS flush on Windows:

```
ipconfig /flushdns
```

- Use stable DNS (for example 8.8.8.8 / 1.1.1.1)

Tesseract OCR not found

- Install Tesseract
- Add to PATH
- Or set `TESSERACT_CMD` in backend `.env`

Frontend cannot reach backend

- Verify backend is running on `http://localhost:5000`
- Verify `VITE_API_URL` in frontend `.env`
- Restart frontend after `.env` changes

11. Development Notes

- Backend type checks were cleaned for key nullable runtime paths in `app.py`.
- Account delete and password change are fully wired backend + frontend.
- Data isolation by `user_email` is applied across stats/history/analytics saves and reads.

12. Security Notes

Current implementation includes bcrypt hashing for password update persistence in backend.

If you plan to deploy publicly, next recommended improvements are:

- Move login/signup validation fully to backend
- Replace local storage auth with token/session-based auth
- Add rate limiting on prediction endpoints
- Add stricter request validation and CSRF protections where needed

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14. License

Educational and academic project.